SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 7.2

Program	**: B. Tech (CSE)**
Specialization	**:**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**II**
Academic Session	**: 2025-2026**
Name of Student	**: N.Dhanush**
Enrollment No.	**: 2403A51L53**
Batch No.	**: 52**
Date	**: 30-01-2026**

Task 1 – Runtime Error Due to Invalid Input Type

(Buggy Code):

```
num = input("Enter a number: ")
result = num + 10
print(result)
```

```
[1]
num = input("Enter a number: ")
result = num + 10
print(result)

Enter a number: 2
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipython-input-1898169331.py in <cell line: 0>()
      1 num = input("Enter a number: ")
----> 2 result = num + 10
      3 print(result)

TypeError: can only concatenate str (not "int") to str

Next steps: Explain error
```

Output:

```
-num = input("Enter a number: ")
+num = int(input("Enter a number: "))
 result = num + 10
 print(result)
```

```
Enter a number: 2
12
```

Task 2 – Incorrect Function Return Value

(Buggy Code):
def square(n):
result = n * n

```
def square(n):
result = n * n

  File "/tmp/ipython-input-3910404483.py", line 2
    result = n * n
    ^
IndentationError: expected an indented block after function definition on line 1

Next steps: Explain error
```

Output:

```
Gemini
def square(n):
-result = n * n
+    result = n * n
```

```
[10]
0s
def square(n):
    result = n * n
```

Task-3 Index Error in List Traversal

(Buggy Code):
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])

```
[11]
0s
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])

  File "/tmp/ipython-input-726334973.py", line 3
    print(numbers[i])
    ^
IndentationError: expected an indented block after 'for' statement on line 2

Next steps: Explain error
```

Output:

```
Gemini
numbers = [10, 20, 30]
-for i in range(0, len(numbers)+1):
-print(numbers[i])
+for i in range(len(numbers)):
+    print(numbers[i])
```

```
10
20
30
```

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Task 4 – Uninitialized Variable Usage

(Buggy Code):
if True:
pass
print(total)

```
[13]
0s
if True:
pass
print(total)

File "/tmp/ipython-input-1170978020.py", line 2
  pass
  ^
IndentationError: expected an indented block after 'if' statement on line 1

Next steps: Explain error
```

Output:

```
Gemini
13]
if True:
-pass
+    pass
print(total)
-
```

```
Gemini
if True:
    pass
+total = 0 # Or any other initial value
print(total)
```

```
0
```

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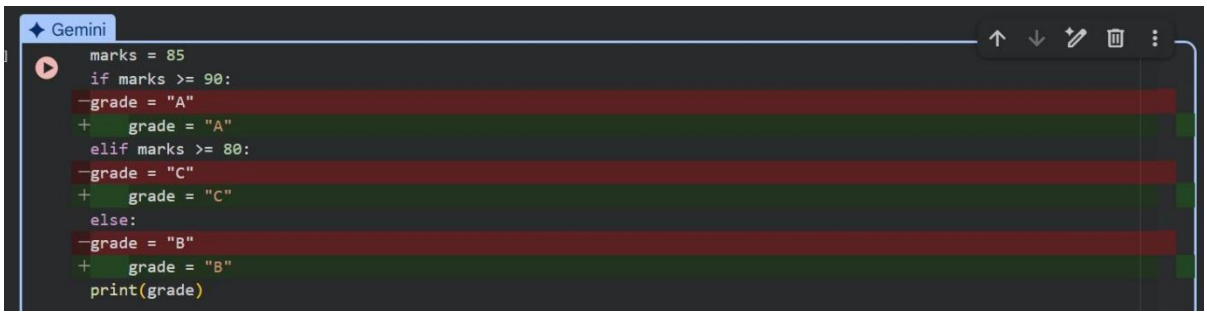
Task 5 – Logical Error in Student Grading System

(Buggy Code):

```
marks = 85
if marks >= 90: grade = "A"
elif marks >=
80: grade = "C"
else:
grade ="B"
print(grade)
```

Output: